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[22AIDEDE11]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER-2023

INTELLIGENT SYSTEMS

Time: 3 Hours Maximum **Marks: 100**

Instructions: Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**05 X 02=10**		
1.	**a.**	Explain graph search	L-1	CO1	**02**
	b.	Write a note on admissibility and consistency	L-2	CO2	**02**
	c.	Write a note on Dijkstra's shortest path algorithm	L-2	CO3	**02**
	d.	Compare finite vs. infinite vs. continuous domains	L-2	CO4	**02**
	e.	Write a note on minimax algorithm	L-2	CO5	**02**
		PART-B	**06 X 05=30**		
2.	**a.**	Explain the term 'Acting humanly'. Also explain the term Turing Test approach?	L-2	CO1	**05**
		OR			
	b.	Define in your own words: **i.** intelligence, **ii.** artificial intelligence, **iii.** agent, **iv.** rationality, **v.** logical reasoning	L-2	CO1	**05**
3.	**a.**	Explain the term Heuristic function with an example. Describe the properties of Heuristic function.	L-2	CO2	**05**
		OR			
	b.	Define Hill climbing algorithm. State the disadvantages of Hill Climbing algorithm.	L-2	CO2	**05**
4.	**a.**	Explain skeletonization in AI with an example	L-2	CO3	**05**
		OR			
	b.	Define Forward State Space Planning and Backward State Space Planning, with an advantage and disadvantage.	L-2	CO3	**05**
5.	**a.**	Explain the following: **i.** Variable Ordering Heuristic **ii.** Value Ordering Heuristic	L-2	CO4	**05**
		OR			
	b.	Explain Finite Constraint Satisfaction Problem (FCSP) with an example.	L-2	CO4	**05**

6.	a.	Briefly explain Games with an element of chance	L-2	CO5	05
		OR			
	b.	Explain alpha - beta pruning with an example.	L-2	CO5	05
7.	a.	Explain with a pseudo code, online search agents.	L-3	CO6	05
		OR			
	b.	Explain Minimax algorithm. Describe the strategies for the same.	L-3	CO6	05
		PART-C	**06 X 10=60**		
8.	a.	Explain the cognitive modeling approach	L-2	CO1	10
		OR			
	b.	Explain with an example the term 'Decision Theory	L-2	CO1	10
9.	a.	Explain simulated annealing with an appropriate example	L-2	CO2	10
		OR			
	b.	Describe the properties of genetic algorithm with an appropriate example.	L-2	CO2	10
10.	a.	Explain the basic issues in Robotics	L-2	CO3	10
		OR			
	b.	Explain Partial Order Planning with an example.	L-2	CO3	10
11.	a.	Describe the three methods to economically benefit the constraint satisfaction problem.	L-2	CO4	10
		OR			
	b.	Explain the extension to quantified constraint satisfaction problem.	L-2	CO4	10
12.	a.	Explain agent environment in AI	L-2	CO5	10
		OR			
	b.	Explain 'poker game' in the perspective of 'games with an element of chance'.	L-2	CO5	10
13.	a.	Explain utility function with an example	L-2	CO6	10
		OR			
	b.	Explain game playing in artificial intelligence.	L-2	CO6	10